

**Section C: Long Essay Questions (LEQ's)****Answer ALL Questions in this section****(40 Marks)**

1. In waste management in the rural setting, pit latrines are commonly used. One of your community member wishes to construct a pit latrine for his family. dimensions,  
Discuss management waste disposal under the following headings; (20 Marks)
  - A. Discuss the information you will provide to the community member so that he can construction a pit latrine of acceptable standard (10 Marks)
  - B. Explain the advantage and disadvantages of a pit latrine (10 Marks)
2. A mother brings a two week old child that she delivered at home for immunization. (20 Marks)
  - A. Explain the services the child will receive at the clinic on that day (10 Marks)
  - B. Describe the complete immunization schedule for this child up to the age of 2 years (10 Marks)







# **MASENO UNIVERSITY**

## **UNIVERSITY EXAMINATIONS 2017/2018**

### **THIRD YEAR END YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (BSc) IN NURSING WITH INFORMATION TECHNOLOGY**

#### **MAIN CAMPUS**

#### **MNS 301: PEDIATRIC NURSING**

Date: 9<sup>th</sup> January, 2018

Time: 9.00 - 12.00pm

#### **INSTRUCTION:**

- Answer ALL Questions
- Answer Questions in the Answer Sheet and booklet provided
- Section A has Thirty (30) Multiple Choice Questions (MCQs) [30]
- Section B has Six (6) Short Answer Questions (SAQs) [30]
- Section C has Two (2) Long Essay Questions (LEQs) [40]



**Section A: Multiple Choice Questions (MCQ's)****(30 Marks)**

Select the most appropriate response to the stem question

1. Prognosis of the acute leukemia of childhood has improved considerably by targeting sanctuary sites where malignant cells may survive. Which of the following is regarded as a "sanctuary site"?
  - A. Muscles and bone
  - B. Skin and mucous membranes
  - C. Testis and the central nervous system
  - D. Pancreas and liver
  - E. Heart
2. All the following statements concerning esophageal atresia are correct, **EXCEPT**:-
  - A. Pregnancy complicated by polyhydramnios should be screened for this anomaly
  - B. It is frequently associated with a tracheo-esophageal fistula
  - C. The problem might be detected during the first feeding, despite a thorough first examination of a newborn
  - D. An operation should be done as soon as possible
  - E. Intubation might be necessary to avoid aspiration
3. Pathophysiologic factors of bronchial asthma include all of the following, **EXCEPT**:-
  - A. Hyper reactivity of the bronchial system
  - B. IgE mediated antigen-antibody reaction
  - C. Genetic predisposition
  - D. Hyperactive sympathetic nervous system
  - E. Formation of mucous plugs
4. Which of the following therapeutic interventions should be carried out first in acute asthmatic attack?
  - A. Increased intake of fluids
  - B. Beta receptor agonists administration
  - C. Beta receptor blockers administration
  - D. Theophylline administration
  - E. Corticosteroids administration
5. The first step in the therapy of a diabetic ketoacidotic coma is:-
  - A. Correction of the acidosis with  $\text{NaHCO}_3$
  - B. Lumbar puncture and analysis of the cerebrospinal fluid
  - C. Fluid replacement with the infusion of an isotonic solution containing
  - D. Subcutaneous administration of insulin
  - E. Fluid replacement with infusion of a glucose-containing solution
6. Which of the following condition is associated with a right to left shunt?
  - A. Patent ductus arteriosus (PDA)
  - B. Atrial septal defect (ASD)

- C. Tricuspid atresia
  - D. Ventricular septal defect (VSD)
  - E. Persistent foramen ovale
7. The most common cause of meningitis in a neonate is:-
- A. Staphylococcus aureus
  - B. Haemophilus influenza
  - C. E-coli
  - D. Meningococcus
  - E. Streptococcus pneumonia
8. The following are guidelines for the therapy of congestive heart failure in a child, **EXCEPT:-**
- A. Bed-rest, half-sitting position
  - B. Administration of oxygen
  - C. Replacement of oral feeding with assisted feeding through naso-gastric tube
  - D. Low salt diet; restricted potassium intake
  - E. Administration of diuretics
9. In a child, the dominant symptom of prolonged presence of a foreign body in the nose is:-
- A. Pain
  - B. Bleeding from the nose
  - C. Purulent nasal discharge
  - D. Lacrimation on the same side
  - E. Obstruction
10. The following symptom is necessary for the diagnosis of diabetes mellitus?
- A. Glycosuria
  - B. Hyperglycemia
  - C. Ketonemia
  - D. Ketonuria
  - E. Metabolic acidosis
11. The following are important aspects of management of a client with bronchiolitis **EXCEPT:**
- A. Supplemental oxygen for infants with hypoxia
  - B. Intravenous fluids and close monitoring of nutritional status
  - C. Good hand washing and hygiene practices
  - D. Antibiotics
  - E. Antipyretics for fever management
12. The clinical manifestation of cyclic fever in *Plasmodium falciparum* associated malaria is caused by :-
- A. Pre-erythrocytic phase
  - B. Hepatic stage
  - C. Erythrocytic stage
  - D. Sexual stage



E. Hypnozoites

13. Before corrective surgery is done, an infant with Coarctation of the Aorta, should be treated with:-

- A. Digoxin
- B. Indomethacin
- C. Prostaglandin E1
- D. Epinephrine
- E. Sildenafil

14. A "tet spell" or "blue" spell of tetralogy of Fallot is treated with all of the following **EXCEPT**:-

- A. Oxygen
- B. Genu pectoral position
- C. Morphine
- D. Digoxin
- E. Propranolol

15. Jones criteria comprises a cluster of symptoms which is diagnostic of:-

- A. Juvenile rheumatoid arthritis
- B. Rheumatic fever
- C. Duchens muscular dystrophy
- D. Hemophilia
- E. Osteomyelitis

16. Which of the following is a developmental milestone in infancy?

- A. Separation Anxiety
- B. Temper tantrums
- C. Sibling rivalry
- D. Object permanence
- E. Toilet training

17. While caring for a child, the nurse notes a pulsating mass in the child's periumbilical area. Which of the following assessments is appropriate for the nurse to perform?

- A. Measure the length of the mass
- B. Auscultate the mass
- C. Percuss the mass
- D. Palpate the mass
- E. Note transillumination characteristics

18. The nurse is assessing a 4 month-old infant. Which motor skill would the nurse anticipate to find in this child?

- A. Hold a rattle
- B. Bang two blocks
- C. Drink from a cup

- D. Wave "bye-bye"
  - E. Clap hands
19. The rationale for administering allopurinol as part of the client's chemotherapy regimen is to:
- A. Prevent metabolic breakdown of xanthine to uric acid
  - B. Prevent uric acid from precipitating in the ureters
  - C. Enhance the production of uric acid to ensure adequate excretion of urine
  - D. Ensure that the chemotherapy doesn't adversely affect the bone marrow
  - E. Prevent destruction of heart muscles
20. Which abnormalities on a full blood count usually suggest acute infection?
- A. Immature neutrophils
  - B. High white cell count
  - C. Thrombocytopenia
  - D. Low mean cell volume
  - F. High plasma viscosity
21. An infant weighed 2.8kgs at birth. If growth occurs at a normal rate, what would be the expected weight at 12 months of age?
- A. Double the birth weight
  - B. Triple the birth weight
  - C. Gain 500 milligrams each month
  - D. Gain 1 gram each month
  - E. Achieved four times its birth weight
22. In Hemophilia A, there is deficiency of factor \_\_\_\_\_
- A. IV
  - B. VII
  - C. VIII
  - D. IX
  - E. X
23. The priority nursing action for a 7 year-old child hospitalized with acute glomerulo-nephritis is:
- A. Assess for generalized edema
  - B. Monitor for increased urinary output
  - C. Encourage rest during hyperactive periods
  - D. Note patterns of increased blood pressure
  - E. Assess for proteinuria
24. A child with diabetic ketoacidosis requires insulin. What lab value do you need to monitor before insulin administration?
- A. Magnesium level
  - B. Sodium level
  - C. Potassium level
  - D. Calcium level
  - E. Chloride level



25. Which of the following is **NOT** an admission criteria for a child with burns?
- A. Burns involving the face
  - B. Burns on limbs across joints
  - C. 40 % total body surface area burns
  - D. Burns in the perineum
  - E. More than two burn accidents in a year
26. In celiac disease:-
- A. There is extensive damage to the large intestine
  - B. There is inflammatory reaction that results in increase B cells which produce cytokines, which cause the intestinal damage
  - C. There is genetic predisposition involving codon 6 of  $\beta$  globin chain
  - D. Amino acid valine is replaced by glutamine
  - E. Children are tolerant gluten protein but intolerant to gliadin
27. Which of the following is a type of secondary lesion?
- A. Macule
  - B. Papule
  - C. Vesicle
  - D. Ulcer
  - E. Hemangioma
28. Emollient cream in a child with eczema serves to:-
- A. Make the skin feel soft
  - B. Prevent evaporation of water from the hydrated epidermis
  - C. Minimize cracking of the dermis
  - D. Prevents inflammation of the skin
  - E. Debride scales produced in the disorder
29. Which age group has the greatest potential to demonstrate regression when they are sick?
- A. Adolescent
  - B. School going age
  - C. Pre-school age
  - D. Toddler
  - E. Infant
30. A four (4) month old infant:-
- A. Uses simple words such as "mama"
  - B. Uses monosyllabic babbling
  - C. Links syllables together
  - D. Coos when comforted
  - E. Has one word with meaning

**Section 2: Short Answer Questions (SAQs)**

**(30 Marks)**

Answer **All** questions in this section

1. Explain five (5) danger signs screened in children when using Integrated Management of Childhood Infections (IMCI) approach. (5 Marks)
2. Explain five(5) factors that affect growth and development of children (5 Marks)
3. Outline six (6) causes of drug and substance abuse among youth in Kenya (6 Marks)
4. Describe pathophysiology of nephrotic syndrome (6 Marks)
5. State four (4) signs of developmental dysplasia of the hips observed in children during physical examination (4 Marks)
6. State and explain any two (2) classifications of Gastroesophageal reflux (GER) (4 Marks)

**Section 3: Long Essay Questions (LEQ's)****(40 Marks)**Answer All Questions in this section.

1. A six month old male patient presents to the emergency department. He was seen 3 days ago with fever and upper respiratory symptoms, diagnosed with otitis media and treated with oral amoxicillin. This morning he became irritable and was less active than usual. He has vomited three times and his urine output is noticeably decreased. He has no diarrhea.

On examination: Temp 40°C, P 90 beats/min, R 30 breaths/min (irregular), BP 120/90, weight 8kg. He is lethargic and arouses only to painful stimuli. His anterior fontanel is full and tense, and he has neck rigidity. His pupils are reactive, but his eyes do not focus well on his parents. His heart, lungs and abdomen are normal. His color and perfusion are good. He has no petechiae. He moves all his extremities weakly and his deep tendon reflexes are hyperactive.

A. What is the most likely medical diagnosis in this client? Give a rationale for your answer. (2 Marks)

B. Identify complications of this disease (2 Marks)

C. Draw a nursing care plan for this client (10 Marks)

D. Explain six (6) aspects of the nervous system evaluated in a pediatric client (6 Marks)

2. A 13 year old female client is admitted with a diagnosis of leukemia. She has been receiving chemotherapy for the past 3. The care-taker reports the client to have had temperature of 38.8°C that did not respond to paracetamol which she has been taking every 8 hours since the previous day, before admission. Her admission assessment indicated clear lung sounds, heart sounds are strong and regular, she is not in distress, has alopecia, and has evidence of white patches in her mouth. She is admitted for further investigations and management

A. Outline five (5) theories that explain childhood neoplasm development (5 Marks)

B. Describe the phases of chemotherapeutic management required in this client, and state the aim of each phase (5 Marks)



- C. Explain five (5) major acute or short term adverse effects of chemotherapy. Give a nursing intervention of how each can be prevented (5 Marks)
- D. Outline how chemotherapeutic agents are classified, based on structure and give an example of a drug in each class (5 Marks)









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### **THIRD YEAR END YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (BSc) IN NURSING WITH INFORMATION TECHNOLOGY**

#### **MAIN CAMPUS**

#### **MNS 303: CLINICAL NUTITION**

Date: 15<sup>th</sup> January, 2018

Time: 2.00 - 5.00pm

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#### **INSTRUCTION:**

- Answer ALL Questions in Sections A, B in the spaces provided
- Section A has Twenty (20) Multiple Choice Questions (MCQs)
- Section B has Twenty (20) Short Answer Questions (SAQs)
- Section C has Thirty (30) Long Essay Questions (LAQs) Answer any TWO Questions in the answer booklet provided



## ANSWER SHEET FOR (TRUE/FALSE) QUESTIONS

REG. NO \_\_\_\_\_

COURSE CODE AND TITLE \_\_\_\_\_

SECTION \_\_\_\_\_

|    | T | F |
|----|---|---|
| 1  |   |   |
| 2  |   |   |
| 3  |   |   |
| 4  |   |   |
| 5  |   |   |
| 6  |   |   |
| 7  |   |   |
| 8  |   |   |
| 9  |   |   |
| 10 |   |   |

### SECTION A (20 MARKS)

**ANSWER ALL QUESTIONS IN THIS SECTION**

**Indicate with a tick, whether the following statements are true/false (10 MARKS)**

**MCQ1.** Pregnancy during adolescence greatly increases needs for iron, calories, and protein.

**MCQ2.** Iron deficiency anaemia is routinely screened for during infancy.

**MCQ3.** Failure to thrive (FTT) is always the result of inadequate dietary intake.

**MCQ4.** HDL favour cholesterol excretion through the synthesis of bile and are referred to as good lipoprotein.

**MCQ5.** The RDA for nutrients generally is the minimum amounts the average adult requires.

**MCQ6.** Monosaccharides are the building blocks on which disaccharides are based.

**MCQ7.** Periconceptional supplementation with folic acid does NOT reduce the incidence of megaloblastic anemia in pregnancy.

**MCQ8.** Food frequency checklist can be used to generate information on diversity and intake.

**MCQ9.** Ageing is a process of slow cell death beginning before fertilization.

**MCQ10.** Zinc deficiency is associated with impaired immune function.

### ANSWER SHEET FOR MCQ QUESTIONS

REG. NO \_\_\_\_\_

COURSE CODE AND TITLE \_\_\_\_\_

SECTION \_\_\_\_\_

|     | a | b | c | d | e |
|-----|---|---|---|---|---|
| 11. |   |   |   |   |   |
| 12. |   |   |   |   |   |
| 13. |   |   |   |   |   |
| 14. |   |   |   |   |   |
| 15. |   |   |   |   |   |
| 16. |   |   |   |   |   |
| 17. |   |   |   |   |   |
| 18. |   |   |   |   |   |
| 19. |   |   |   |   |   |
| 20. |   |   |   |   |   |

**Choose the correct answer(s) that apply to each of the following questions**

**(10 marks)**

**MCQ11.** Which of the following statements best describes dietary standards?

- A. They are guidelines that help us understand the nutrient intake of a healthy human being.
- B. It is a daily nutrient intake value that is estimated to meet the requirement of half of the healthy individuals in a life stage and gender group
- C. It is the average daily dietary intake level that is sufficient to meet the nutrient requirement of nearly all (97 to 98 percent) healthy individuals in a particular life stage and gender group.
- D. They are guidelines that help us understand how much of a particular nutrient is needed by a healthy human being.
- E. None of the above

**MCQ12.** Which of the following amino acids is the odd one out?

- A. Threonine
- B. Glycine
- C. Alanine
- D. Arginine
- E. Asparagine

**MCQ13.** Energy Expenditure for basal metabolism is:

- A. The energy expended when resting, but awake and in fasting state
- B. The energy expended when doing moderate activity
- C. The energy expended to make food available to the body
- D. The energy expended when sleeping
- E. The energy expended when sleeping and in fasting state

**MCQ14.** Nutrient density can be defined as

- A. The amount of kcalories in a food divided by the amount of kcalories needed in a day
- B. The amount of a particular nutrient in a serving of food divided by the number of kcalories in that serving.
- C. The amount of a particular nutrient in a serving of food divided by the number of grams of protein.
- D. Nutrient needed for the day
- E. Kcalories needed for the day

**MCQ15.** Most digestion takes place in the:

- A. Pancrease
- B. Small intestine
- C. Large intestine
- D. Stomach
- E. Mouth

**MCQ16.** Which of the following is not an organic nutrient?

- A. Carbohydrates
- B. Proteins
- C. Minerals
- D. Vitamins
- E. All of the above

**MCQ17.** Extracellular fluid is approximately \_\_\_\_\_ of body weight

- A. 10%
- B. 20%
- C. 30%
- D. 40%
- E. 50%

**MCQ18.** Non-heme iron is provided from:

- A. Plant sources and elemental iron components of plants structures
- B. Animal sources and elemental iron components of animal tissue
- C. Animal flesh
- D. Herbal plants

E. None of the above

**MCQ19.** The BMI indices for a healthy human being is:

- A.  $<18.5 \text{ kg/m}^2$
- B.  $>30 \text{ kg/m}^2$
- C.  $18.5\text{-}24.9 \text{ kg/m}^2$
- D.  $25\text{-}29.9 \text{ kg/m}^2$
- E.  $16\text{-}25 \text{ kg/m}^2$

**MCQ20.** All of the following are indicators of assessing nutritional status except:

- A. Weight/height
- B. Nutrient intake
- C. Physical signs
- D. Head circumference
- E. All of the above

### SECTION B (20 MARKS)

**SAQ21.** State **FOUR (4)** roles of fats in the human body. (2 marks)

**SAQ22.** Define the following terms. (2 marks)

(a) Health

(b) Essential nutrients

**SAQ23.** State **TWO (2)** factors that influence our food choices. (2 marks)

**SAQ24.** Mention **TWO (2)** uses of the RDA. (2 marks)

**SAQ25.** Indicate whether the following amino acids are essential amino acids or non-essential amino acids by placing 'EAA' for essential or 'NEAA' or non-essential in the appropriate space: (5pts)

|       |               |
|-------|---------------|
| _____ | Valine        |
| _____ | Histidine     |
| _____ | Asparagine    |
| _____ | Glutamine     |
| _____ | Isoleucine    |
| _____ | Glycine       |
| _____ | Arginine      |
| _____ | Glutamic acid |
| _____ | Serine        |
| _____ | Proline       |

**SAQ26.** Indicate scientific names of the following: (3½ marks)

\_\_\_\_\_ Vitamin A

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Vitamin B<sub>6</sub>  
 Vitamin B<sub>7</sub>  
 Vitamin E  
 Vitamin K  
 Vitamin B<sub>5</sub>  
 Vitamin B<sub>9</sub>

**SAQ27.** Identify the deficiency disease or the cause of the deficiency disease as required in the table below: (3½ marks)

| Deficient vitamin | Deficiency disease                         |
|-------------------|--------------------------------------------|
| Retinol           |                                            |
|                   | Neural tube defects                        |
| Phylloquinone     |                                            |
|                   | Pernicious anaemia                         |
| Niacin            |                                            |
|                   | Rickets in children/Osteomalacia in adults |
| Tocopherols       |                                            |

### SECTION C (30 MARKS)

Each question in this section carries **FIFTEEN (15) MARKS**. Answer any **TWO (2)** questions in the answer booklet provided.

**LAQ28.** (a) State the **TWO (2)** classifications of dietary fibres and in each case give an example. (3 marks)

(b) Discuss **THREE (3)** positive and **THREE (3)** negative effects of dietary fibres on health. (12 marks)

**LAQ29.** (a) Explain the difference between complete and incomplete proteins and in each case give **FOUR (4)** examples. (6 marks)

(b) State **THREE (3)** factors that affect protein requirements. (3 marks)

(c) One of the functions of proteins is formation of essential body components: Explain the **THREE (3)** components. (6 marks)

**LAQ30.** (a) Discuss **FOUR (4)** common nutritional related problems associated with pregnancy. (8 marks)

(b) Explain the advantages of breastfeeding over other forms of infant feeding.

(7 marks)



